

JEE MAIN REVISION MASTER: APPLICATION OF DERIVATIVES

Subject: Mathematics | Target Scope: JEE Main Previous Years | Empirical Base: 33 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Very High	Increasing/Decreasing Functions & Maxima / Minima	21 PYQs	Optimization
High	Rate of Change, Tangents & Normals	12 PYQs	Derivatives Applications

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
Increasing/Decreasing Functions & Maxima / Minima	$\text{Increasing: } f'(x) \geq 0, \quad \text{Decreasing: } f'(x) \leq 0, \quad \text{Maxima: } f'(c)=0 \text{ and } f''(c)<0$
Rate of Change, Tangents & Normals	$\text{Slope of Tangent: } m = \frac{dy}{dx}, \quad \text{Slope of Normal: } m_n = -\frac{1}{dy/dx}, \quad m_1 m_2 = -1$

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Increasing/Decreasing Functions & Maxima / Minima (Very High Yield — 21 PYQs)

Core Principle: Monotonicity: $f(x)$ is strictly increasing in (a, b) if $f'(x) > 0$; strictly decreasing if $f'(x) < 0$. Maxima/Minima: Critical points where $f'(x) = 0$. Second derivative test: If $f'(c) = 0$ and $f''(c) < 0$, then $x=c$ is a point of local maximum; if $f''(c) > 0$, it is a local minimum. If $f''(c) = 0$, test fails \implies use first derivative test.

■ **Caution / JEE Pitfall:** Absolute maximum and minimum of a continuous function on a closed interval $[a, b]$ can occur at CRITICAL POINTS or at the ENDPOINTS a and b !

■ **Speed Trick:** To find intervals of increase/decrease, find roots of $f'(x) = 0$ and apply the wavy curve method on the number line.

Representative Source PYQs:

- [2026 Shift] Q#1 (MCQ): "A YouTube short video is getting viral according to $f(t)=-2t^3+3t^2+5$. At what time does the video get maximum number of shares? (\$t...\$)" [Ans: N/A]
- [2025 Shift] Q#3 (MCQ): "The function $f(x)=\tan x-x...$ " [Ans: N/A]
- [2024 Shift] Q#1 (MCQ): "If $f(x)=xe^x(1-x)$, then $f(x)$ is..." [Ans: N/A]

3.2 Rate of Change, Tangents & Normals (High Yield — 12 PYQs)

Core Principle: Rate of change $(dy/dt) = (dy/dx)(dx/dt)$. Slope of tangent to $y=f(x)$ at (x_1, y_1) is $m = f'(x_1)$; equation is $y - y_1 = m(x - x_1)$. Slope of normal is $-1/m$; equation is $y - y_1 = -(1/m)(x - x_1)$. Two curves cut orthogonally if $m_1 m_2 = -1$.

■ **Caution / JEE Pitfall:** When normal is parallel to x-axis, its slope is 0, which means $(dx/dy) = 0$ or $(dy/dx) \rightarrow \infty$!

■ **Speed Trick:** For sphere: $V = (4/3)\pi r^3$ implies $(dV/dt) = 4\pi r^2 (dr/dt) = S (dr/dt)$.

Representative Source PYQs:

- **[2026 Shift] Q#2 (MCQ):** "In a Mahakumbh, a drone camera is moving along $3y=x^3-3$. When y -coordinate changes 9 times as fast as x -coordinate, it captures..." [Ans: C]
- **[2024 Shift] Q#1 (MCQ):** "The value of C in (0,2) satisfying the mean value theorem for the function $f(x)=x(x-1)^2, x \in [0,2]$ is equal to..." [Ans: N/A]
- **[2023 Shift] Q#2 (MCQ):** "A circular plate of radius 5cm is heated. Due to expansion, its radius increase at the rate of 0.05cm/s. The rate at which its area is ..." [Ans: N/A]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: KCET 2026 (Q#1)

A YouTube short video is getting viral according to $f(t) = -2t^3 + 3t^2 + 5$. At what time does the video get maximum number of shares? (t is in hours)

- (A) 1
- (B) 2
- (C) 3
- (D) 4

Correct Answer: None

Solution Analysis: The number of shares is given by $f(t) = -2t^3 + 3t^2 + 5$. To find when the video gets the maximum number of shares, we differentiate $f(t)$ with respect to t . $f'(t) = (d/dt)(-2t^3 + 3t^2 + 5) = -6t^2 + 6t$. For maximum or minimum value, put $f'(t) = 0$. $-6t^2 + 6t = 0$. $6t(1-t) = 0$. So, $t = 0$ or $t = 1$. Now check the nature using the second derivative. $f''(t) = -12t + 6$. At $t = 1$, $f''(1) = -12(1) + 6 = -6$. Since this is negative, $f(t)$ has a maximum at $t = 1$. So, the video gets the maximum number of shares after 1 hour. Correct option: Option A.

Benchmark Problem 2: KCET 2026 (Q#2)

In a Mahakumbh, a drone camera is moving along $3y = x^3 - 3$. When y -coordinate changes 9 times as fast as x -coordinate, it captures good quality pictures. Then one of the precise positions of the drone at that instant is

- (A) (-3, -8)
- (B) (3, -8)
- (C) (3, 8) [Correct]
- (D) (-3, 8)

Correct Answer: C

Solution Analysis: The path of the drone is given by $3y = x^3 - 3$. The statement says that the y -coordinate changes 9 times as fast as the x -coordinate. This means $(dy/dx) = 9$. Now differentiate the given equation with respect to x : $3y = x^3 - 3$ $3(dy/dx) = 3x^2$. So, $(dy/dx) = x^2$. At the required instant, $x^2 = 9$. $x = \pm 3$. Now check the points on the curve. For $x = 3$: $3y = 3^3 - 3$. $3y = 27 - 3 = 24$. $y = 8$. So one position is (3, 8). Hence the correct option is Option C (3, 8).